



X816/76/01

Computing Science

Marking Instructions

Please note that these marking instructions have not been standardised based on candidate responses. You may therefore need to agree within your centre how to consistently mark an item if a candidate response is not covered by the marking instructions.

General marking principles for Higher Computing Science

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

- (a) Marks for each candidate response must **always** be assigned in line with these general marking principles and the detailed marking instructions for this assessment.
- (b) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted.
- (c) If a candidate response is not covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.
- (d) Award marks regardless of spelling, as long as the meaning is unambiguous. This applies to all responses, including code. Award marks as per the detailed marking instructions, regardless of syntax errors, if the intention of the coding is clear.
- (e) For questions where candidates are asked to design or write code, a sample response is shown in the detailed marking instructions. This will not be the only valid response. You must use the detailed marking instructions and additional guidance to ensure that you consider alternative approaches and nuances of different programming languages. If in doubt you should refer to your Team Leader.
- (f) A correct response can be negated if the candidate includes an extra, incorrect response. For example, if the candidate is asked for two answers for two marks and the candidate gives three, one of which is incorrect, they are awarded one mark.
- (g) If a candidate scores through a response and makes a further attempt, you should only mark the further attempt. If no further attempt is made and the original is legible, you should mark the original response.
- (h) Where an incorrect response is carried forward and used correctly in a following part of the question, you should give credit for subsequent responses that are correct with regard to the original error. Candidates should not be penalised more than once for the same error.
- (i) Only award marks for a valid response to the question asked. Where candidates are asked to:
 - **Identify, name, give or state**, they need only name or present in brief form.
 - **describe**, they must provide a statement or structure of characteristics and/or features. This will be more than an outline or a list. It may refer to, for example, a concept, process, experiment, situation, or facts, in the context of and appropriate to the question. Candidates must make the same number of factual/appropriate points as there are marks available in the question.
 - **explain**, they must relate cause and/or effect and/or make relationships between things clear, in the context of the question or a specific area within the question.
 - **write code**, they must write recognisable code, not prose nor a diagram.
 - **design**, they must use a design technique appropriate to the problem. Award marks as per the detailed marking instructions, regardless of errors in the exemplification of the technique, if the intention of the design is clear.
- (j) In the detailed marking instructions, if a word is underlined then it is essential; if a word is bracketed() then it is not essential. Words separated by / are alternatives

Marking instructions for each question

Section 1 – Software design and development and computer systems

Question			Expected response	Max mark	Additional guidance
1.	(a)		1111 0111	1	
	(b)		-128 to 127 OR -2^7 to 2^7-1	2	
2.	(a)		- increase the number of cores - increase width of data bus - add cache/increase cache - any other valid response	1	1 mark for any one bullet
	(b)		Matching explanation - multiple instructions simultaneously - more bits transferred in a single operation - reduces number of accesses to slower main memory - any other valid response	1	1 mark for bullet that matches method stated in part (a) Do not award the mark for increasing clock speed as it is in the stem.
3.			- autonomous driving (1) is more fuel efficient due to system controlling accelerating/ decelerating and detecting/anticipating braking (1) - intelligent route planning (1) reduces driving time by monitoring external factors such as accidents/volume of traffic which reduces fuel consumption (1) - tracking parking (1) reduces driving time searching for space and therefore fuel consumption (1) - engine management system (1) optimises engine efficiency reducing fuel consumption (1) - intelligent road traffic management systems (1) adjusting speed limits to optimise traffic flow reducing fuel consumption (1) - any other valid response	2	1 mark for feature 1 mark for justification related to fuel

Question			Expected response	Max mark	Additional guidance
4.			• Sign bit: 1 • Remaining 15 bits of mantissa: 110 0000 0000 0000 • Exponent: 1111 1110	3	1 mark for each bullet
5.			• authenticates the sender • guarantees the integrity of the sent item/sent item has not been altered	2	1 mark for each bullet OR Award second mark if candidate describes mechanism for integrity or authenticity. For example • generate a code/hash for the document and when received the code/hash is recalculated and compared OR • code/hash is encrypted with sender's private key and decrypted using the public key
6.	(a)		• Condition for length of string • Condition for count of valid characters	2	1 mark for each bullet Example answers <code>IF lengthCheck =true AND countValid=6 THEN</code> <code>IF length(dob)=6 and countValid=6 THEN</code>
	(b)		• checks characters even if the length <>6 • keeps checking characters even when an invalid character is found	1	1 mark for any one bullet Do not accept generic answers relating to efficiency. Answer must be in context and refer to the code.

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	- record structure with name - four fields specified	2	1 mark for each bullet Record structure could be Type, Structure, Class, Named Tuple, Record name IS Example Answers: SQARL RECORD competitorData IS (STRING name, STRING club, STRING event, REAL distance) VB Private/Public/Structure Type competitorData name as string club as string event as string distance as single END TYPE Livecode Global CompetitorDATA Put empty into CompetitorData[1][name] Put empty into CompetitorData[1][club] Put empty into CompetitorData[1][event] Put empty into CompetitorData[1][distance] Python Class method class Competitor (): def __init__(self): self.name = "" self.club = "" self.event = "" self.distance = 0.0 Python Dataclass method Class Competitor: name: str = "" club: str = "" event: str = "" distance: float = 0.0

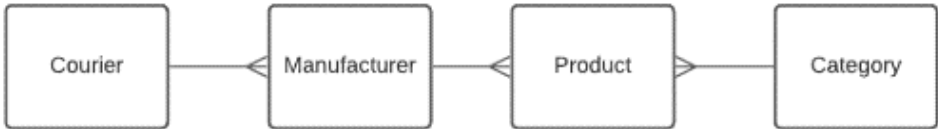
Question			Expected response	Max mark	Additional guidance
7.	(a)	(ii)	- array structure - using data type created in (i)	2	1 mark for each bullet Example Answers: Python <code>competitors=[datatype_a() for x in range (800)]</code> VB <code>DIM competitors(799) AS datatype_a</code> Livecode Set the data of eventData to CompetitorData
	(b)		- open and close file - loop traversing the array with termination - IF with condition event="Javelin" - AND condition distance>=70 - writing name and club to file	5	1 mark for each bullet Open file Loop for all 800/to end of array If event="Javelin" AND distance>=70 then Write name, club to file End if End loop Close file

Question		Expected response	Max mark	Additional guidance
7.	(c)	- ask for searchName/searchEvent and their use in the IF statement - loop that traverses correctly with termination IF condition with: - correct use of array variable - comparison of current name and event fields to targets - display if found with correctly concatenated message - display if not found using any suitable flag, for example -1 or found is false Example answers: <pre> RECEIVE searchName FROM KEYBOARD RECEIVE searchEVENT FROM KEYBOARD DECLARE found INITIALLY false SET counter TO -1 REPEAT counter=counter+1 IF searchName=competitors[counter].name AND searchEvent=searchName=competitors[counter].event then SET found TO true END IF UNTIL found=true OR counter=799 IF found =true THEN SEND competitors[counter].name & "threw " &competitors[counter].distance & "in the " & competitors[counter].event TO DISPLAY ELSE SEND " Competitor not found" TO DISPLAY END IF </pre> Using a fixed loop: <pre> RECEIVE searchName FROM KEYBOARD RECEIVE searchEVENT FROM KEYBOARD DECLARE found INITIALLY false FOR index FROM 0 TO 799 DO IF searchName=competitors[index].name AND searchEvent=competitors[index].event then SEND competitors[index].name & "threw " & competitors[index].distance & "in the " & competitors[index].event TO DISPLAY SET found TO true END IF END FOR IF found =false THEN SEND 'Competitor not found' TO DISPLAY END IF </pre>	6	1 mark for each bullet Candidates' solutions will vary in terms of efficiency but marks are awarded for any correct implementation.

Question			Expected response	Max mark	Additional guidance										
8.	(a)		STEP 2 IN: password STEP 3 IN: randPos1, randPos2, randPos3	2	1 mark for each step										
	(b)		Returns a value	1											
	(c)		The data flow informs parameter passing.	1	The mark is for identifying the relationship to parameter passing. Candidates that use actual or formal incorrectly here should not be penalised as part (d) assesses the use of formal and actual parameters.										
	(d)	(i)	- correct identification of any formal parameter - corresponding actual parameter	2	1 mark for each bullet <table><tr><th>Formal</th><th>Actual</th></tr><tr><td>word</td><td>password</td></tr><tr><td>first</td><td>ranPos1</td></tr><tr><td>second</td><td>ranPos2</td></tr><tr><td>third</td><td>ranPos3</td></tr></table>	Formal	Actual	word	password	first	ranPos1	second	ranPos2	third	ranPos3
Formal	Actual														
word	password														
first	ranPos1														
second	ranPos2														
third	ranPos3														
		(ii)	Local / only in the getLetters sub-program	1	Award mark if candidate references appropriate line numbers.										
		(iii)	- use of conditional loop - use of comparison to number(s) already generated	2	1 mark for each bullet Candidates may choose to exemplify through the use of design or code, for example. Do SET second TO <random number...> Loop until second<>first Do Set third TO <random number...> Loop until third<>second AND third<>first										
	(e)		- allows fewer lines of code/removes repetition (1) by calling same procedure more than once (with different actual parameters) (1) - local variables can use less memory (1) as they only allocated memory during the running of the procedure / memory space is freed at the end of the procedure (1)	2	1 mark for reason which relates to procedures. 1 mark for justification. Do not penalise candidates that refer to functions.										

Question			Expected response	Max mark	Additional guidance
9.	(a)		- count number of matching restaurants - find highest rated restaurant	2	1 mark for each bullet
	(b)		- initialising total and incrementing total by 1 - loop that traverses array with termination IF statement with - condition for food type AND - condition for city	4	1 mark for each bullet Pseudocode example: (Get targetFood from user Get targetCity from user) Set count to 0 Start loop for each restaurant If restaurantType[index] = targetFood AND city[index] = targetCity Set count to count +1 End if End loop
	(c)	(i)	- A=4.12 - B=4.99 - C=false	3	1 mark for each bullet
		(ii)	- a breakpoint will halt execution of the code at a predefined point (various line numbers are suitable) - then the values of variable can be inspected to compare with trace table values/expected values	2	1 mark for each bullet Award second mark if candidates refer to stepping through the code to compare with trace table values as this is another expression of second bullet point.
		(iii)	Change > to < OR swap highest and list[index]	1	
	(d)		- function called correctly with parameter - assigning result to myHighest	2	1 mark for each bullet SET myHighest TO findMax (rating)

Section 2 – Database design and development

Question			Expected response	Max mark	Additional guidance										
10.			See entity-relationship diagram below.	3	1 mark for each relationship - Courier - Manufacturer (1:M) - Manufacturer - Product (1:M) - Product - Category (M:1)										
															
11.			- grouping on rating - corresponding values for each rating - sort in descending order	3	1 mark for each bullet <table border="1"><thead><tr><th>rating</th><th>Longest movie</th></tr></thead><tbody><tr><td>PG</td><td>143</td></tr><tr><td>12A</td><td>130</td></tr><tr><td>15</td><td>119</td></tr></tbody></table>	rating	Longest movie	PG	143	12A	130	15	119		
rating	Longest movie														
PG	143														
12A	130														
15	119														
12.	(a)		Field(s) and Calculation(s) - Use of both type and alias [Most Expensive Item] - Maximum aggregate function on the price field Search Criteria - Use of complex condition / wildcard Grouping - type	4	1 mark for each bullet Search criteria can also use a wildcard, for example type LIKE “*bag” Allow use of % for wildcard. Allow use of ‘maximum’ in the design of a query.										
			Example answer: <table border="1"><tbody><tr><td>Field(s) and calculation(s)</td><td>type, MAX(price) AS [Most Expensive Item]</td></tr><tr><td>Tables(s) and query</td><td>Snack</td></tr><tr><td>Search criteria</td><td>WHERE type = "Multi-bag" or type="Family bag"</td></tr><tr><td>Grouping</td><td>type</td></tr><tr><td>Sort order</td><td></td></tr></tbody></table>			Field(s) and calculation(s)	type, MAX(price) AS [Most Expensive Item]	Tables(s) and query	Snack	Search criteria	WHERE type = "Multi-bag" or type="Family bag"	Grouping	type	Sort order	
Field(s) and calculation(s)	type, MAX(price) AS [Most Expensive Item]														
Tables(s) and query	Snack														
Search criteria	WHERE type = "Multi-bag" or type="Family bag"														
Grouping	type														
Sort order															

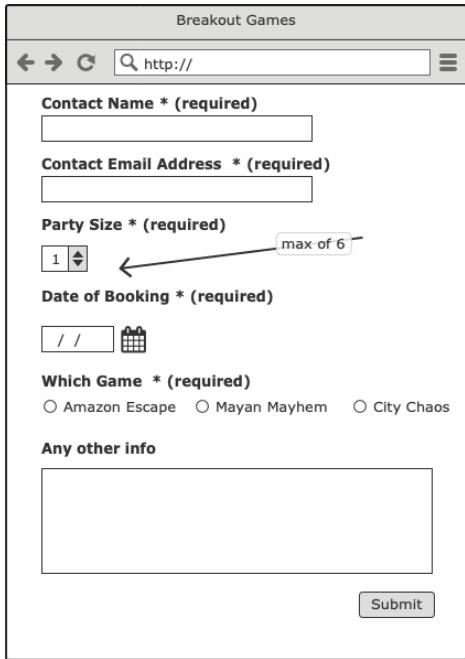
Question			Expected response	Max mark	Additional guidance
12.	(b)		• UPDATE Snack table • SET price to new value (price*0.5 OR price/2) • WHERE criteria based on stock level	3	1 mark for each bullet Example answer: <pre>UPDATE Snack SET price = 0.5*price WHERE stock > 500</pre>
	(c)		• SELECT clause with type and SUM(stock) • Alias as [Total Stock] with FROM and GROUP BY unchanged • ORDER BY with sum(stock) in descending order	3	1 mark for each bullet Example answer: <pre>SELECT type, sum(stock) AS [Total Stock] FROM Snack GROUP BY type ORDER BY sum(stock) DESC</pre>
	(d)		• DELETE FROM Snack • WHERE criteria on product name uses both wildcards	2	1 mark for each bullet Example answers: <pre>DELETE FROM Snack WHERE productName LIKE "*Salt*"; Delete FROM Snack WHERE productName LIKE "%Salt%";</pre> Note <code>"*salt*"</code> or <code>"%salt%"</code> is allowed.

Question			Expected response	Max mark	Additional guidance										
13.	(a)		A customer may rent the same item more than once (which would not be unique).	1											
	(b)		Field(s) and Calculation(s) - Use of three fields customerID, itemType and startDate, and the calculated field (noOfDays * dailyRentalPrice) Tables - Two tables i.e. Rental, Items Search Criteria - Use of complex condition OR of wildcard for July 2020 dates Sort order - startDate with correct sort order Example answer <table><tr><td>Field(s) and calculation(s)</td><td>customerID, itemType, startDate, (noOfDays * dailyRentalPrice)</td></tr><tr><td>Tables(s) and query</td><td>Rental, Item</td></tr><tr><td>Search criteria</td><td>AND startDate LIKE "*/07/2020" OR AND startDate LIKE "%/07/2020" (AND Item.itemID = Rental.itemID AND Customer.customerID = Rental.customerID)</td></tr><tr><td>Grouping</td><td></td></tr><tr><td>Sort order</td><td>startDate ASC</td></tr></table>	Field(s) and calculation(s)	customerID, itemType, startDate, (noOfDays * dailyRentalPrice)	Tables(s) and query	Rental, Item	Search criteria	AND startDate LIKE "*/07/2020" OR AND startDate LIKE "%/07/2020" (AND Item.itemID = Rental.itemID AND Customer.customerID = Rental.customerID)	Grouping		Sort order	startDate ASC	4	1 mark for each bullet Equi-join not needed in design. Search criteria could be expressed as startDate>="01/07/2020" AND startDate<="31/07/2020" Allow use of # around date. As SQL defaults to sorting by ascending order, ASCs does not need to be specified.
Field(s) and calculation(s)	customerID, itemType, startDate, (noOfDays * dailyRentalPrice)														
Tables(s) and query	Rental, Item														
Search criteria	AND startDate LIKE "*/07/2020" OR AND startDate LIKE "%/07/2020" (AND Item.itemID = Rental.itemID AND Customer.customerID = Rental.customerID)														
Grouping															
Sort order	startDate ASC														
	(c)		- FROM clause includes Item and MaxRental - WHERE clause is dailyRentalPrice = [Max daily rental]	2	Example answer <pre>SELECT itemType, itemModel FROM Item, MaxRental WHERE dailyRentalPrice = [Max daily rental];</pre>										

Section 3 - Web design and development

Question			Expected response	Max mark	Additional guidance
14.	(a)		Appropriate function name with parameter in brackets and subsequent use of parameter for colour change to black	1	Example answer <pre>function displayBlack(my_text) { my_text.style.color="Black" ; }</pre>
	(b)		- Add onmouseout event to element - Match function name to name created in part (a)	2	1 mark for each bullet Example answer <pre><p onmouseover="displayRed(thi s)" onmouseout=" displayBlack (this)" > Sale now on</p></pre>
15.			- Grouping of floorplan and event to width 700px - Grouping gaminglogo and eventlogo to width/height 75 px, float right and margin-right 20 px - Implementing the 10 px margins in any of the grouping selectors	3	1 mark for each bullet Example answers: <pre>Header,nav,footer { margin- left 10px;margin-right: 10px;} #gaminglogo,eventlogo { width: 75px; height: 75px; float:right; margin-right: 20px;} #event,#floorplans {width: 700px; margin-left: 10px; }</pre> OR <pre>Header, nav, footer{margin- right:10px} Header, #event, footer, nav, #floorplan{margin- left:10px} #gaminglogo, #eventlogo{width:75px; height:75px; float:right; margin-right:20px} #floorplan, #event{Width:700px}</pre>

Question			Expected response	Max mark	Additional guidance
16.	(a)	(i)	Submit (button)	1	
		(ii)	<code><input type="submit"></code>	1	Additional attributes for the input are not required.
	(b)	(i)	Checked	1	
		(ii)	<code><select name="instrument" size="3" multiple></code>	1	
	(c)		• persona may be that of a non-technical, reading ability, accessibility issue • allows developer to assess/observe/gain feedback on aspects of the website for example navigation, use of form, visual layout etc.	2	Do not accept young, elderly or age on its own.
	(d)		• the heading is a block element • which takes up all of the width of its container element • changing it to an inline element allows the heading and the image to take up the width it requires	2	1 mark for each bullet Maximum 2 marks

Question			Expected response	Max mark	Additional guidance
17.	(a)		<pre>.eventimage { margin-top: 10px; margin-left: 10px ; margin-right: 20px ;}</pre> OR <pre>.eventimage { padding-top: 10px; padding -left: 10px ; padding -right: 20px ;}</pre>	2	All 3 properties for 2 marks any 2 properties for 1 mark
	(b)		• submit button And • text fields with validation indicated e,g, required or use of * for contact name, email and also with text area for additional information • date with validation, for example date picker or dd/mm/yyyy or similar • numeric field with validation for party size i.e. max 6 • radio buttons/select element (drop-down) for game theme	4	1 mark for submit AND 1 mark for each subsequent bullet to a maximum of three. 
	(c)		• developer benefit - changes based on feedback can be made by the developer before development or implementation time has been wasted • client benefit - can provide feedback on matching their requirements/fitness for purpose and suggest changes	2	1 mark for each bullet
	(d)	(i)	<pre>nav ul {list-style-type:none}</pre>	1	
		(ii)	• use of the descendant selector • only selects unordered lists within the nav element/doesn't affect other unordered lists not within a nav element	2	1 mark for each bullet

[END OF MARKING INSTRUCTIONS]